

DATABASE SYSTEMS – Assignment 1 Documentation

Introduction

The assignment was primarily focused around Entity Relationship Diagrams (ERDs) and Enhanced Entity Relationship Diagrams (EERDs) in which case studies were provided and their diagrams had to be made accordingly.

Software Tool Used

For making ERDs and EERDs, draw.io was used. This Diagramming Software tool provided a friendly user interface with easy access to shapes and objects. It does not require subscriptions or any payments.

Assumptions & Clarifications

There were few assumptions made while attempting to draw the diagram of questions:

Question Number 1

1. Every Customer will have a shopping cart.

Question Number 2

1. Since it is not stated whether one teacher will teach only one class or multiple classes, it is assumed that teacher will teach only one class.
2. Participation is made in such a way that not all members enroll in classes and not all instructors take classes. Similarly, not all classes have members enrolled in it.

Question Number 3

1. It has been supposed that a technician will be the one doing maintenance of the ATM Machine, as he is the one hired for this job.
2. The ATM maintenance record has been kept independent and not relied totally on technicians, because even if technicians do not exist, the ATM machine would maintain the maintenance records (Record it as 'Empty' if no records).
3. It is assumed that more than one ATM machine can exist in a bank branch.

Question Number 4

1. The artworks have been categorized based on type of art.
2. A new type of art has been added which is Carving.
3. Honoree does not entirely depend on sculpture, because a list of honorees will also be maintained at the museum. So even if a sculpture is removed the honoree information will stay there, so it is an independent entity.
4. Artworks are also categorized according to their status or whether they are permanent in the museum or they are borrowed.

Question Number 5

1. Both Undergrads and grads can become Teaching Assistant that's why Union is used between it.
2. Both the Graduate Assistant and Teacher's Assistant are Non-Tenured Faculty.

3. The students are divided into initial subclasses based on degree. There is further partial division based on roles and responsibilities of some students.

Apparent Constraints

There are many constraints that could not be represented using ERD. These include:

1. **CHECKS:** Means what values attributes can or cannot take. Eg, Transaction type in Question 3 can only take three possible values namely "Withdrawal", "Deposit" or "Transfer". Further, it may include checks like height of an artwork in Question 4 cannot be negative or zero.
Also in Question 3, customers will not be able to make any transaction unless they have at least one bank account and card.
2. **General Constraints:** It includes providing NULL, NOT NULL in different scenarios. For instance, primary keys cannot be NULL. But there are some cases where NULL is necessary and has meaning like in Question 5, faculty who do not drive a car on campus may be represented as NULL.
Also in Question 4, the death dates are NOT NULL, if the person is alive.